



The past, present, and future of brand research

Travis Tae Oh¹  · Kevin Lane Keller² · Scott A. Neslin² · David J. Reibstein³ · Donald R. Lehmann⁴

Published online: 04 June 2020

© Springer Science+Business Media, LLC, part of Springer Nature 2020

Abstract

This article discusses the past, present, and future of brand research. We begin by reviewing three historical eras of branding development in the past: the *information*, *attribute*, and *brand equity eras*. Each era is marked by an emphasis on a particular concept of brands and a dominant research methodology during that period. We characterize the present of brand research as the *branding era*, in which brands have become part of the everyday vocabulary, and the concept of “branding” has been embraced by all types of organizations and people. Finally, we share our perspective on the future of brand research by discussing three major shifts in the brand landscape, mostly influenced by technological advancements such as IoT and their capacity to collect granular data at the individual level. We anticipate that new, exciting opportunities in brand research will continue to open up at the intersection of brands, technologies, and people.

Keywords Brands · Branding · Brand research history · Brands and technology

✉ Travis Tae Oh
to2280@columbia.edu

Kevin Lane Keller
kevin.keller@dartmouth.edu

Scott A. Neslin
scott.a.neslin@tuck.dartmouth.edu

David J. Reibstein
reibstein@wharton.upenn.edu

Donald R. Lehmann
drl2@columbia.edu

Extended author information available on the last page of the article

1 Introduction

Brands are instrumental to marketers and consumers alike. For centuries, brands have served as effective carriers of functional and symbolic information about goods, services, and experiences, allowing customers to efficiently associate personal meaning to a company's products. It is now almost impossible to avoid exposure to brands, and such ubiquity constantly shapes people's thoughts and behaviors even without their conscious awareness. Indeed, the success of a company in today's marketplace is largely attributable to the power of its brand—i.e., the level of awareness, strength of associations, and overall brand attitudes it engenders. These far-reaching implications have spurred much academic attention on brands since the early days of marketing research (e.g., Gardner and Levy 1955), which continues to be of great interest for both practitioners and academics.

This article provides a high-level discussion of the past, present, and future of brand research. The primary purpose is to examine major trends and propose future research directions, rather than review each subfield of brand research in detail¹. First, we identify three areas of focus that developed chronologically in the past: *the information era*, *the attribute era*, and *the brand equity era*. We then discuss current research, which we call the *branding era*. The emphasis is on creating and developing brands. We identify five noteworthy brand research areas, and highlight a few key findings. Lastly, we offer our take on future research topics on brands in light of technological advancements and macro changes in the market.

2 Brand research in the past (1970s–early 2000s)

2.1 The information era

In the Indus Valley, circa 2000 B.C., a cow meandered off to a greener pasture to be with other cows. How could the owner retrieve the cow and prove it was his cow? Brand it. The farmer took a branding iron and burned his farm's insignia onto the cow. Now, the cow easily could be identified, no matter where it wandered off to. The Old Norse word for “to burn” is “brandr.”² Thus, the initial role of brands was to associate the product with its producer. Over the years, pottery makers, brick makers, paper makers, or anyone producing something that needed to be identified jumped on the “brandwagon.”

The industrial revolution raised this to a higher level. Ford Motor Company, for instance, could now produce for the entire United States and internationally as well. But how could consumers in California know the car they were buying in fact was the real thing—a Ford? The answer was to physically identify the brand; the Ford logo graced the Model T, assuring consumers they were buying the right vehicle.

Hence, the initial role of brands was to alleviate consumer uncertainty and risk by associating the brand with a given producer. Economists in the 80s formalized this

¹ For a detailed review, see Keller and Lehmann (2006) or Avery and Keinan's (2015) chapter on “Consuming Brands”

² See <https://www.skyword.com/contentstandard/creativity/branding-brief-history>

notion, launching the “information era” of brand research. Shapiro (1983) showed how reputation, attributed to the product by knowing its brand, resolved uncertainty and allowed the firm to charge higher prices. Erdem and Swait (1998) used survey data to show that “When consumers are uncertain about product attributes, firms may use brands . . . to ensure that their product claims are credible.” Wernerfelt (1988) showed how the *brand signal* could be extended to assure the quality of new products. The brand as an identifier of the producer became nuanced—a *signal* of brand quality.

2.2 The attribute era

Eventually, researchers and practitioners started to explore how to create high-quality brands. This led to a focus on product *attributes* (e.g., convenience, reliability, taste, nutrition), and gave rise to one of the most important concepts in marketing—the value proposition: Crest toothpaste prevents cavities; Excedrin relieves headaches; Polaroid produces a picture in one minute.

Marketers initially treated attributes as tangible—for example, the size of a monitor. But soon marketers acknowledged that attributes could also be intangible. This entailed images and associations, ranging from prestigious (e.g., an “Ivy League” institution) to cultural (e.g., Pepsi is “for the Pepsi generation” or Chevrolet is about “baseball, hot dogs, apple pie and Chevrolet”). Abstract branding messages proliferated during this era. In its seminal 1984 Super Bowl ad, Apple established itself as revolutionary, different, and innovative. Researchers began to incorporate “image” in brand research (Dobni and Zinkhan 1990; Padgett and Allen 1997).

In the ’70s and ’80s, the emphasis was mostly on tangible attributes. Attribute-based linear attitude models and conjoint analyses dominated brand research. Rooted in the Fishbein model of attitude (Fishbein 1967), Bass and Talarzyk (1972) predicted brand preference as a function of attribute ratings, while Lehmann and O’Shaughnessy (1974) investigated the relative importance of various attributes in determining preference. A vibrant debate centered on whether, in fact, all attributes were equally important (Beckwith and Lehmann 1973). Conjoint analysis proved to be insightful in this regard, showing how much of one attribute consumers were willing to trade off to get a certain amount of another attribute (Green and Rao 1971; Green and Wind 1975).

As the field progressed, researchers related attributes not only to stated preference but to actual brand choice, utilizing the multinomial logit model (Guadagni and Little 1983). For example, Gensch (1984) applied the logit model to predict supplier choice in a B2B context, using attributes such as reliability, price, ease of installation, and warranty. Similarly, Kamakura and Russell (1993) predicted choice based on both tangible and intangible (i.e., image) attributes. Erdem et al. (2005) showed consumers learn product quality by tapping into multiple information sources along the path to purchase.

The focus on attributes was academically important and managerially relevant. Yet, results from conjoint studies repeatedly suggested that brands were not only about attributes, be they tangible or intangible. For example, Green and Wind (1975) showed significant brand identification effects in conjoint models, even after controlling for several important attributes. Though it is possible that the conjoint models were not accounting for enough attributes, researchers concurred that something was missing—brand equity.

2.3 The brand equity era

The first challenge was to define brand equity. Several researchers stepped in with varying definitions (see Table 1). These diverse definitions share commonalities. First, the brand name of a product conveys something extra compared to an unbranded product or one with a different name. Second, brand equity has concrete implications such as brand volume, margins, response to marketing, and other measurable outcomes. The consensus across the definition is that brand equity translates brand identification into added value for the brand.

The next task was to measure brand equity. Ailawadi et al. (2003) identify three categories of brand equity measures:

- *Customer mindset*: Measures such as awareness, attitude, attachment, or loyalty. Mindset measures have been commercialized by approaches from companies such as Young and Rubicam's Brand Asset Valuator
- *Product-market outcomes*: Measures based on outcomes such as market share (Chaudhuri and Holbrook 2018), price premium (Agarwal and Rao 1996), profit (Dubin 1998; Goldfarb et al. 2009), and revenue premium (Ailawadi et al. 2003)
- *Financial market outcomes*: Measures that attempt to quantify the brand name as a financial asset (Simon and Sullivan 1993; Mizik 2014)

Subsequent studies in branding quantified concrete causes and effects of brand equity. For example, Ailawadi et al. (2003) measured brand equity as the added revenue a branded product received relative to an unbranded product. They found that brands in high-priced product categories with long inter-purchase times enjoyed higher brand equity. One of their most important findings was that brand equity is "sticky." The equity a brand garners in the current year closely reflects its equity in the previous year. Managerial actions such as advertising could improve or maintain equity but brand equity is generally difficult to build, although not to destroy, in a short timeframe.

Table 1 Sample definitions of brand equity

Researcher	Definition
Farquhar (1989)	"We define brand equity as the 'added value' with which a given brand endows a product"
Aaker (1991)	"A set of brand assets and liabilities linked to a brand, its name and symbol, that add to or subtract from the value provided by a product or service to a firm and/or to that firm's customers"
Keller (1993)	"The differential effect that brand knowledge has on consumer response to the marketing of that brand"
Keller (2002)	"The set of associations and behaviors on the part of the brand's customers . . . that permits the brand to earn greater volume or greater margins than it could without the brand name"
Ailawadi et al. (2003)	"Brand equity is defined as the marketing effects or outcomes that accrue to a product with its brand name compared with those that would accrue if the same product did not have a brand name"

In summary, initial research on brands took us through three eras: brands as producer information, brands as attributes, and brands as equity. Importantly, the stage was set for elaboration and extension of these ideas.

3 Brand research in the present: the branding era (2000s–2020)

Although brands have been around for centuries and marketers have recognized the importance of brand equity for many years, the research environment and developments over the past two decades can be characterized as the *branding* era. Brands have become part of the everyday vocabulary, and the concept of “branding” has been embraced by all types of organizations and people. Coinciding with this trend is a deeper and broader exploration of brands and branding by practitioners and academics alike. In terms of the latter, the word “brand” now appears in the title of at least one article in virtually every issue of a major marketing journal.

Academics have employed a variety of different methodologies, from survey or experimental techniques (Aaker 1997; Thomson et al. 2005) to more qualitative approaches (Fournier 1998; Muniz and O’Guinn 2001). This modern era of branding is marked by a study of a myriad of different research topics. Though many topics have been studied by academics during this time, we point out five broad branding areas for their noteworthy insights and promise for the purposes of this article. For each area, we identify three to four important topics and highlight some recent key findings on one of those topics in greater detail.

3.1 Noteworthy areas of current study in branding

Brand fundamentals Three topics of continued interest in branding are (1) naming and logo design criteria; (2) financial impact and value of brands; and (3) brand equity dimensions. For example, analyses of the four brand equity dimensions as measured by the aforementioned Brand Asset Valuator suggest that *differentiation* often performs differently from *relevance*, *esteem*, or *knowledge*. Making a brand more distinctive than other brands may also make the brand less familiar and well-regarded because of greater uncertainty of its meaning:

- “The authors uncover a fairly strong positive association of sales-based brand equity (SBBE) with three dimensions of customer-based brand equity (CBBE)—relevance, esteem, and knowledge—but a slight negative correspondence with the fourth dimension, differentiation.” (Datta et al. 2017)
- “Unlike the other three dimensions, differentiation is negatively associated with customer acquisition and retention, although positively associated with profit margin.” (Stahl et al. 2012)

Brand intangibles Four topics of continued interest in understanding brand intangibles are (1) brand authenticity; (2) brand feelings and emotions; (3) brand attachment and community; and (4) brand personality and anthropomorphism. Researchers are probing the different ways that consumers experience, engage, or

even advocate for brands and the effects that are created as a result. In terms of the fourth topic, research continues to explore human-like characteristics of brands through brand personality traits but also investigates specific brands that deliberately take on anthropomorphism. For example:

- “Brand anthropomorphization increases the perceived unfairness of price increases and the perceived fairness of price decreases.” (Kwak et al. 2015)
- “Consumers’ inferences of an anthropomorphized brand's intentionality to interact with them in a socially crowded context trigger greater social withdrawal, thereby resulting in lower preferences for the brand.” (Puzakova and Kwak 2017)
- “Materialists respond more favorably to a servant brand than to a partner brand when the brand is anthropomorphized (vs. objectified), and they respond more favorably to an anthropomorphized servant brand than do nonmaterialists.” (Kim and Kramer 2015)

Marketing effects in a digital era Four topics of prime interest in understanding marketing effects in a digital era are (1) social media and word-of-mouth; (2) opinion leaders and influencers; (3) extracting brand meaning from online data; and (4) integration with traditional media. Research in these areas has demonstrated how the consumer journey leading up to purchase and consumption has shifted in profound ways in a digital era. Yet, in terms of the fourth topic, research has also validated the continued effectiveness of more traditional media, at least with today's consumer, as follows:

- “Thoroughly orchestrating traditional advertising and a firm's social media activities may improve a firm's performance with respect to building the brand and encouraging customer acquisition” (de Vries et al. 2017).
- “Overall, we find that earned and paid media play a central role in developing and maintaining entertainment brands” (Lovett and Staelin 2016).
- “The results indicate that companies benefit from using social media (e.g., Twitter) for personalized customer responses, although there is still a role for traditional brand communications, such as press releases and advertising (Hewett et al. 2016).

Brand architecture Three topics of continued interest in understanding brand architecture are (1) brand extensions and dilution; (2) corporate image; and (3) co-branding and brand alliances. In terms of the third topic, research has shown that brand partnerships can be helpful only under certain conditions:

- “A partner brand's social media power potential, power exertion, and their interaction can all lead to higher composite product sales.” (Kupfer et al. 2018)
- “When product benefit information is presented simultaneously with co-branding information (at a delay after co-branding information), the presence of a well-known brand weakened (strengthened) the association of the less-known brand with the outcome and thereby lowered (improved) evaluation of the less-known brand.” (Cunha et al. 2015)

The role of brands Three topics of continued interest in understanding the role of brands are (1) cultural differences; (2) legal consequences; and (3) brands as symbols. Understanding how brands acquire different meaning and how that is manifested in different domains is of fundamental importance. In terms of the third topic, research shows that the meaning and power of brands can be transformative in multiple circumstances:

- “Brand use can enhance feelings of self-efficacy, which can lead to better task performance.” (Park and John [2014](#))
- “Consumption of a performance-branded product improves objective outcomes in a variety of contexts, although consumers attribute gains to themselves.” (Garvey et al. [2016](#))
- “Close brand relationships can help insulate consumers against physical pain due to brands’ ability to provide a semblance of social connectedness.” (Reimann et al. [2017](#))

3.2 Current status of branding research

Brands are a pervasive presence in consumers’ lives and central to marketplace exchange and social interaction. While we have learned much about brands and branding in recent years, research continues to reveal different facets of brands and their direct and indirect effects. One clear take-away from the last two decades of research, as illustrated above, is that brands are rich in meaning and complex in their effects. One consequence of this is that research going forward must recognize and appreciate the large number of concepts and factors that potentially come into play as mediators and moderators of branding effects.

4 Brands in the future: what’s next? (~ 2030s)

We explore how the brand landscape may look in the not too distant future by discussing how technological shifts and market changes may impact the role of brands in the marketplace, and how we study them. Specifically, we identify three major interconnected areas that we expect to be pertinent to the future of brands and brand research: (1) technological advancement and its impact on the role and visibility of brands, (2) an extended scope of brand research, beyond products and companies, and (3) heightened capacities of automated data collection and analyses leading to changes in branding strategies. We also present specific research questions for each of the areas.

4.1 Technology and the role of brands

Perhaps the biggest evolution affecting brands is the influence of technology on brands and how customers interact with their brands. The Internet of things (IoT) will change how products and brands relate to their customers. As products follow our movements, calories burned and consumed, how and where we travel, and with whom we interact—brands will have much more information about their customers.

This will enable companies to further customize and shape their offerings, communications, and connections with the customers, which may solidify the relationships between customers and their brands. Yet, much is to be learned about the potential barriers that IoT may create for new brands and acquiring new customers. Furthermore, it is unclear how technologies such as artificial intelligence, virtual reality, and augmented reality will change the relationships customers have with their brands, especially in regards to how these technologies will be further improved from what we see now (see Lee et al. 2020 for more on this issue).

With rapidly improving features of virtual reality (“VR”) and augmented reality (“AR”) that have been adopted by major online marketplace platforms (e.g., Amazon), we expect that the experiential gap between online and offline vendors will be even further reduced. Because online stores are less subject to space constraints, smaller brands have become more readily available to customers. For instance, a search of Bluetooth headphones on [Amazon.com](https://www.amazon.com) shows more than 40,000 results, whereas a brick and mortar retailer could not physically carry 40,000 SKUs for Bluetooth headphones. We predict that the concentration of sales among the dominant brands may dissipate while that among online retailers may concentrate, but the question remains as to which brands in which industries will be less or more influenced by this shift that is being accelerated by developments in AR and VR.

Future research questions in this area include (1) How will advanced forms of IoT and AR/VR influence customers’ relationships with brands? (2) Will enhanced behavioral information gathered from IoT increase barriers to entry for new brands? If so, which industries will be more or less affected? (3) How will advanced AR/VR technology influence customer-brand interactions in the online and offline marketplace? Will it lead to increased or decreased power of dominant brands? (4) Will online retailers need to maintain a bricks-and-mortar presence to nurture its brand (cf. Amazon, Bonobos, Warby Parker)?

4.2 Extended scope of brands: beyond products and companies

Considerable research on brands has focused on branding of products and services. However, geographic areas (nations, regions, and cities) as well as individuals will also establish their brands. For example, New York City, London, or Paris are brands. Individual experts, journalists, and politicians are also brands. Measuring the value of geographic brands and their economic consequences has not been fully explored beyond country of origin (e.g., Piron 2000; Lampert and Jaffe 1998). A country or geographical region as a brand has also been explored, most notably by Anholt (1998). Yet, only recently, have efforts emerged to examine the economic consequences of the brand of a country³. The need for understanding the economic consequences of a nation’s brand or how brand campaigns lead to a country’s brand image are still quite understudied.

Similarly, much more research can be conducted for brands of individual persons. This is best understood through celebrity endorsements of brands, but considerable work remains to be done on this front as well (see Keel and Natarajan 2012). More and more “non-celebrity” individuals are building their own brands through independent media channels such as YouTube or Twitch,

³ <https://www.usnews.com/news/best-countries/overall-rankings>

while very little has been studied on the empirical and theoretical implications of individuals as brands.

Another extension of brand research is its role *within* the organization (Aurand et al. 2005). The value of a brand has often been characterized using measures generated by consumer demand (product-market outcomes as noted earlier). What has been largely overlooked is the impact of the brand from the human resources side—one critical issue being the ability to retain talent. Given the major financial implications of employee turnover and efficiency, internal branding is of great value to the organization and has not yet been sufficiently quantified.

Thus, future research questions include (1) What are the economic consequences of geographical regions as brands? (2) What are the key antecedents to building a brand for geographical regions, such as countries? (3) What are the implications and processes of building a brand for an individual? (4) What are the implications of internal branding on human resource management?

4.3 Advancements in data collection and analysis

With the advent of Big Data and improved ability to understand specific needs of individual customers through IoT, companies are not only producing products unique to one individual's preferences but also providing unique messages, promotions, and prices catered to the individual. Highly advanced virtual audio assistants from Amazon, Apple, and others are figuring out what you want based on your conversations before you login to do a search. With precise geo-tracking information, companies now make offers based on customers' unique locations. The rapidly expanding means of individual-level data collection and analyses, however, comes with major implications for consumers, brands, and society. Customized marketing efforts raise concerns regarding privacy and inequality, which have already incited much debate and even led to "opt-in" regulations regarding personal information⁴. More research is needed to determine the role of brands in this new environment, especially in regards to consumers' experiences—the sensory, affective, intellectual, and behavioral interactions (Brakus et al. 2009)—with brands that sell IoT products.

Another possible scenario of in the 2030s, as technologies on individual-level data collection proliferate, is that brands as we know it today may disappear. Highly advanced artificial intelligence and machine learning systems may simply scour the choices available and make selections on behalf of customers, who no longer need to rely on brands as a signal of quality or features. It is highly likely that this automation will dominate customer-firm interactions, as also discussed by Oblander et al. (2020) in this special issue. Future research could empirically examine when and how brands influence the customer's journey, especially when customers are mostly represented by machine algorithms.

Future research questions thus include (1) What role does brand image play in encouraging customers to share personal information? (2) Is it desirable (for the brand, the customer, or society in general) to encourage sharing of personal information? (3) How do brand experiences and relationships change in highly automated search and selection processes?

⁴ For example, obtaining cookies data now requires permission ("opt-in") from customers in the EU. <https://gdpr.eu/cookies/>

5 Conclusion

More than seven decades of brand research have broadened our understanding of how brands affect various aspects of a firm's performance (e.g., margins, revenue) by influencing consumers' attitudes and behaviors. As a core marketing concept, brand research has traversed multiple research paradigms over time, namely the *information*, *attributes*, and *brand equity* eras. Brands initially served as a means of identification to assure customers the source and quality of products. Soon, however, practitioners and academics realized that brands convey intangible symbolic meaning along with tangible product attributes. Research on quantifying and measuring the effect of its attributes (e.g., via conjoint analysis) eventually led to the conceptualization of brand equity. The accumulating empirical evidence of brand equity's significant impact on firm performance fueled greater research interest on how and when brands affect consumers' thoughts, feelings, and behaviors.

This past research left the field with a heightened recognition of the psychology of brands, triggering the *branding era* of the present. From brand anthropomorphization to complexities in co-branding efforts, current research in branding spans across a wide range of topics, including five notable research areas: *brand fundamentals*, *brand intangibles*, *marketing effects in a digital era*, *brand architecture*, and *the role of brands*. We expect that while branding research will continue to thrive as it relates to the aforementioned topics and others, it may also face paradigmatic shifts arising from macro changes in consumer demand, mainly due to technological developments.

In the dawning era of advanced technology, we anticipate that the role and influence of brands will dramatically change. For one, the growing dominance of the online marketplace, along with AR and VR developments, will fundamentally alter the way products are distributed and purchased. Furthermore, we expect that the abundance of individual-level data from online platforms and IoT will trigger major changes in how brands build their relationships with their customers, and possibly lead to mass customizations at all levels of industries. Lastly, we anticipate that mass data collection and analyses at the individual level will engender major questions on the role of brands in consumer privacy and fairness. Managers now face a more complex and automated process of brand communications, and would benefit from more empirical research on how brands influence the customer's journey in such contexts.

For centuries, brands have served a quintessential role in marketing. Though certain aspects of branding, such as attributes or brand images, were emphasized and researched at different time periods, brands have and will continue to be an important topic of study for marketing academics and practitioners. We anticipate that new, exciting opportunities in brand research will continue to open up at the intersection of brands, technologies, and people.

References

- Aaker, D. A. (1991). *Managing brand equity*. New York: The Free Press.
- Aaker, J. L. (1997). Dimensions of brand personality. *Journal of Marketing Research*, 34(3), 347–357.
- Agarwal, M. K., & Rao, V. R. (1996). An empirical comparison of consumer-based measures of brand equity. *Marketing Letters*, 7(3), 237–247.

- Ailawadi, K. L., Lehmann, D. R., & Neslin, S. A. (2003). Revenue premium as an outcome measure of brand equity. *Journal of Marketing*, 67(10), 1–17.
- Anholt, S. (1998). Nation-brands of the twenty-first century. *Journal of Brand Management*, 5, 395–406.
- Aurang, T. W., Gorchels, L., & Bishop, T. R. (2005). Human resource management's role in internal branding: An opportunity for cross-functional brand message synergy. *Journal of Product and Brand Management*, 14(3), 163–169.
- Avery, J., & Keinan, A. (2015). Consuming brands. In M. I. Norton, D. D. Rucker, & C. Lamberton (Eds.), *The Cambridge handbook of consumer psychology*. New York: Cambridge University Press.
- Bass, F. M., & Talarzyk, W. W. (1972). An attitude model for the study of brand preference. *Journal of Marketing Research*, 9(1), 93–96.
- Beckwith, N. E., & Lehmann, D. R. (1973). The importance of differential weights in multiple attribute models of consumer attitude. *Journal of Marketing Research*, 10(2), 141–145.
- Brakus, J. J., Schmitt, B. H., & Zarantonello, L. (2009). Brand experience: What is it? How is it measured? Does it affect loyalty? *Journal of Marketing*, 73(5), 52–68.
- Chaudhuri, A., & Holbrook, M. B. (2018). The chain of effects from brand trust and brand affect to brand performance: The role of brand loyalty. *Journal of Marketing*, 65(2), 81–93.
- Cunha, M., Forehand, M. R., & Angle, J. W. (2015). Riding coattails: When co-branding helps versus hurts less-known brands. *Journal of Consumer Research*, 41(2), 1284–1300.
- Datta, H., Ailawadi, K. L., & van Heerde, H. J. (2017). How well does consumer-based brand equity align with sales-based brand equity and marketing-mix response? *Journal of Marketing*, 81(5), 1–20.
- de Vries, L., Gensler, S., & Leeflang, P. S. H. (2017). Effects of traditional advertising and social messages on brand-building metrics and customer acquisition. *Journal of Marketing*, 81(9), 1–15.
- Dobni, D., & Zinkhan, G. M. (1990). In search of brand image: A foundation analysis. *Advances in Consumer Research*, 17, 110–119.
- Dubin, J. A. (1998). The demand for branded and unbranded products: An econometric method for valuing intangible assets. Chapter 4. In *Studies in consumer demand: Econometric methods applied to market data* (pp. 77–127). Norwell: Kluwer Academic Publishers.
- Erdem, T., & Swait, J. (1998). Brand equity as a signaling phenomenon. *Journal of Consumer Psychology*, 7(2), 131–157.
- Erdem, T., Keane, M. P., Oncu, T. S., & Strebel, J. (2005). Learning about computers: An analysis of information search and technology choice. *Quantitative Marketing and Economics*, 3(3), 207–246.
- Farquhar, P. H. (1989). Managing brand equity. *Marketing Research*, 1(9), 24–33.
- Fishbein, M. (1967). In M. Fishbein (Ed.), *Readings in attitude theory and measurement*. New York: John Wiley and Sons.
- Fournier, S. (1998). Consumers and their brands: Developing relationship theory in consumer research. *Journal of Consumer Research*, 24(4), 343–373.
- Gardner, B., & Levy, S. (1955). The product and the brand. *Harvard Business Review*, 33(3–4), 33–39.
- Garvey, A. M., Germann, F., & Bolton, L. E. (2016). Performance brand placebos: How brands improve performance and consumers take the credit. *Journal of Consumer Research*, 42(4), 931–951.
- Gensch, D. H. (1984). Targeting the switchable industrial customer. *Marketing Science*, 3(1), 41–54.
- Goldfarb, A., Lu, Q., & Moorthy, S. (2009). Measuring brand value in an equilibrium framework. *Marketing Science*, 28(1), 69–86.
- Green, P. E., & Rao, V. R. (1971). Conjoint measurement for quantifying judgmental data. *Journal of Marketing Research*, 8(8), 355–363.
- Green, P. E., & Wind, Y. (1975). New way to measure consumers' judgments. In *Harvard Business Review* (Vol. 7–8, pp. 107–117).
- Guadagni, P. M., & Little, J. D. C. (1983). A logit model of brand choice calibrated on scanner data. *Marketing Science*, 2(Summer), 203–238.
- Hewett, K., Rand, W., Rust, R. T., & van Heerde, H. J. (2016). Brand buzz in the echoverse. *Journal of Marketing*, 80(5), 1–24.
- Kamakura, W. A., & Russell, G. J. (1993). Measuring brand value with scanner data. *International Journal of Research in Marketing*, 10, 9–22.
- Keel, A., & Natarajan, R. (2012). Celebrity endorsement and beyond: New avenues for celebrity branding. *Psychology and Marketing*, 29(9), 690–703.
- Keller, K. L. (1993). Conceptualizing, measuring, and managing customer-based brand equity. *Journal of Marketing*, 57(1), 1–22.
- Keller, K. L. (2002). *Branding and brand equity, MSI Relevant Knowledge Series*. Cambridge: Marketing Science Institute.

- Keller, K. L., & Lehmann, D. R. (2006). Brands and branding: Research findings and future priorities. *Marketing Science*, 25(6), 740–759.
- Kim, H. C., & Kramer, T. (2015). Do materialists prefer the brand-as-servant? The interactive effect of anthropomorphized brand roles and materialism on consumer responses. *Journal of Consumer Research*, 42(8), 284–299.
- Kupfer, A. K., vor der Holte, N. P., Kübler, R. V., & Hennig-Thurau, T. (2018). The role of the partner brand's social media power in brand alliances. *Journal of Marketing*, 82(5), 25–44.
- Kwak, H., Puzakova, M., & Rocereto, J. F. (2015). Better not smile at the price: The differential role of brand anthropomorphization on perceived price fairness. *Journal of Marketing*, 79(7), 56–76.
- Lampert, S. I., & Jaffe, E. D. (1998). A dynamic approach to country-of-origin effect. *European Journal of Marketing*, 32(1/2), 61–78.
- Lee, B. C., Moorman, C., Moreau, C. P., Stephen, A. T., & Lehmann, D. R. (2020). Research on innovation: The past, present, and future. *Marketing Letters*.
- Lehmann, D. R., & O'Shaughnessy, J. (1974). Difference in attribute importance for different industrial products. *Journal of Marketing*, 38(2), 36–42.
- Lovett, M. J., & Staelin, R. (2016). The role of paid, earned, and owned media in building entertainment brands: Reminding, informing, and enhancing enjoyment. *Marketing Science*, 35(1-2), 142–157.
- Mizik, N. (2014). Assessing the total financial performance impact of brand equity with limited time-series data. *Journal of Marketing Research*, 51(6), 691–706.
- Muniz, A. M., & O'Guinn, T. C. (2001). Brand community. *Journal of Consumer Research*, 27, 412–432.
- Oblander, E. S., Gupta, S., Mela, C. F., Winer, R. S., & Lehmann, D. R. (2020). Research on innovation: The past, present, and future. *Marketing Letters*.
- Padgett, D., & Allen, D. (1997). Communicating experiences: A narrative approach to creating service brand image. *Journal of Advertising*, 26(4), 49–52.
- Park, J. K., & John, D. R. (2014). I think I can, I think I can: Brand use, self-efficacy, and performance. *Journal of Marketing Research*, 51(4), 233–247.
- Piron, F. (2000). Consumers' perceptions of the country-of-origin effect on purchasing intentions of (in)conspicuous products. *Journal of Consumer Marketing*, 17(4), 308–321.
- Puzakova, M., & Kwak, H. (2017). Should anthropomorphized brands engage customers? The impact of social crowding on brand preferences. *Journal of Marketing*, 81(11), 99–115.
- Reimann, M., Nuñez, S., & Castaño, R. (2017). Brand-aid. *Journal of Consumer Research*, 44(10), 673–691.
- Shapiro, C. (1983). Premiums for high quality products as returns to reputations. *The Quarterly Journal of Economics*, 98(4), 659–680.
- Simon, C. J., & Sullivan, M. W. (1993). The measurement and determinants of brand equity. A financial approach. *Marketing Science*, 12(Winter), 28–52.
- Stahl, F., Heitmann, M., Lehmann, D. R., & Neslin, S. A. (2012). The impact of brand equity on customer acquisition, retention, and profit margin. *Journal of Marketing*, 76(7), 44–63.
- Thomson, M., MacInnis, D. J., & Park, C. W. (2005). The ties that bind: Measuring the strength of consumers' emotional attachments to brands. *Journal of Consumer Psychology*, 15(1), 77–91.
- Wernerfelt, B. (1988). Umbrella branding as a signal of new product quality: An example of signaling by posting a bond. *The RAND Journal of Economics*, 19, 458–466.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Affiliations

Travis Tae Oh¹ · Kevin Lane Keller² · Scott A. Neslin² · David J. Reibstein³ · Donald R. Lehmann⁴

¹ Columbia Business School, Columbia University, New York, NY 10027, USA

² Tuck School of Business, Dartmouth College, Hanover, NH 03755, USA

³ Wharton School of the University of Pennsylvania, Philadelphia, PA 19104, USA

⁴ Columbia Business School, Columbia University, New York, NY 10027, USA